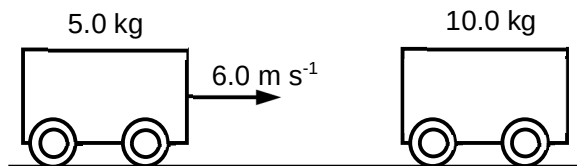


- 5 A trolley of mass 5.0 kg travelling at a speed of 6.0 m s^{-1} collides head-on and locks together with another trolley of mass 10.0 kg which is initially at rest. The collision lasts for 0.20 s.



What is the total kinetic energy of the two trolleys after the collision and the average force acting on each trolley during this collision?

	Total kinetic energy after the collision / J	Average force on each trolley / N
A	30	150
B	75	150
C	30	100
D	75	100

Applying conservation of linear momentum, the speed after the collision is 2.0 m s^{-1} .

Hence, the total kinetic energy after the collision is $\frac{1}{2} m_{\text{total}} v^2 = \frac{1}{2} \times 15 \times 2.0^2 = 30 \text{ J}$.

The force on either trolley is $\Delta p / \Delta t = 20 / 0.20 = 100 \text{ N}$.

6 Three blocks with masses M , $2M$ and $3M$ are pushed along a rough surface by a horizontal force